Remember That Time? Telling Interesting Stories from Past Interactions

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Abstract. Sociability is a human trait that plays a central part in relationships over time. Today, humans are increasingly in long-term interactions with intelligent agents, which have proven most useful when they are sociable. Such sociability requires the agent to remember and appropriately refer to past interactions. A common way in which humans refer to their past interactions and collaborations is through storytelling. Such stories, often abbreviated, include a small set of interesting and extraordinary events. We propose the design, development and preliminary evaluation of a generic computational architecture for finding and retelling such interesting event sequences. Our system mines interesting interaction episodes in a corpus of prior interactions. Initial evaluation of interactions selected by the system for retelling are encouraging. A future goal of the research is to support collaborative composition of stories about prior interactions between humans and agents in a mixed-initiative framework to produce interesting retellings.

Keywords: Social interaction · Storytelling · Story generation · Human-robot interaction · Narrative content selection

1 Introduction

In the rapidly growing category of interactive robots and virtual agents, sociability seems increasingly important in establishing and maintaining an effective interaction or long-term relationship. There is also an increasing amount of data of interactions that is being recorded on individual players within and across games. Importantly, the utility of a particular episode of such interaction is not limited to the conclusion of the tasks defined for it. We, as humans, reflect upon some such episodes in our daily interactions with each other as memorable points that can reshape a relationship's dynamics. This might be because the value of some social activities, such as "watching a sports game together", often goes beyond the main task itself, and so an interesting sub-sequence of it could be memorable and worthy of retelling.

Similarly, the sociability of an agent plays a meaningful and often integral role in fulfilling the agent's tasks. For instance, the human-robot interaction community has been working on the notion of sociability for robots and virtual

agents and has underlined its impact in numerous studies and in different contexts [7]. However, the memorable moments in interacting with these agents are not necessarily the tasks or sub-tasks themselves, but most often the interesting events that happen during them, which can relate or even be quite unrelated to the main context of interaction.

Hence, it can be very rewarding for an agent to be able to construct compelling retellings of personalized interaction episodes that are considered interesting. Such story retellings can potentially cause higher levels of user engagement through their emotional connection with a more sociable agent or game character [3].

From another point of view, the success of an interaction or collaboration with an agent in any context can be measured in various objective and subjective ways, and from different perspectives. For instance, context-based evaluations can assess the performance of the task in which the agent has been helping a user, or emotional analyses can estimate the levels of frustration, boredom or happiness and engagement. But intuitively, stories are our way of describing a fun, useful, healthy or otherwise negative and bothersome experience. Ideally, a good long-term interaction with a social agent, or a fun gameplay with a social game character, over time, should be interesting enough so that the user is compelled to tell or hear a story about some memorable episodes of it. For example, there are popular web-logs and social network pages dedicated to telling interesting stories that personal assistants currently shipped in smartphones trigger the users to retell.

Note that we are specifically talking about social agents, and it is perfectly normal, for example, if a utility robot (e.g., doing pizza delivery or house cleaning) does not trigger the users to say any stories; although interestingly enough, many users currently using such robots do have stories to tell about them through anthropomorphizing them and imagination (e.g. Roomba robots [8]).

This work focuses on the first aspect of the problem which is to find *interesting* events from past interactions. There is another aspect which is in the way any set of past events is *communicated*. With richer text generation toolkits, it is possible to make the agents recall even a mundane set of events but communicate them in an interesting way. The focus for this paper is a framework for representation of interactions and selection and annotation of interesting events. Future work will address richer natural language and visual composition of co-constructed stories from these selected events.

In this paper, we will first describe a software architecture and approach to annotate and record generalized gameplay interactions. We define and use the markup language notation AIL (Abstract Interaction Logs) and an algorithm that uses sequential rule mining to extract interesting events across multiple interactions in a game of *rummy*. Our current results are based on the choice of algorithm and a heuristic to find interesting events, but the methodology and implementation is expandable to any set of interactions marked up in AIML and a generic or game-specific heuristic.

2 Related Work

Chess has been a particularly popular domain for exploring story generation possibilities among many researchers. The most recent and relevant works in this area include [5] and [11]. Buchthal et al. suggest an interesting system capable of importing a chess game and applying its extracted features to one of their four available story "skins", to retell them based on the dramatic elements of the chess game. This work does not produce stories directly from the game interactions, but rather uses its extracted features to alter other skins.

Pablo Gervás takes a different approach to generating stories from chess games. Their work is more related to this paper, in that it tries to find interesting event sequences in a chess game. It does so by selecting and scoring the best "focalized" experiences within the game, which are the chosen as representatives of an overall story. Events are described relative to game pieces and their fields of vision. Moreover, each game piece is considered as a story character. Gervás' approach to finding compelling event sequences is geared toward better summarization. We focus on a general notion of interestingness rather than saliency. We also do not require recalling entire games but rather single events within past games.

Additionally, both works by Buchthal et al. and Gervás differ in a few fundamental ways from what we are considering. Not only they are limited to chess as a domain, they only consider the abstract game data as an input, and treat it as a series of isolated events that are separated from a possible interaction between the characters during which the game might have happened. In contrast, we are also interested in considering other types of events and actions that can occur inside a game of chess, or any other context of interaction.

Another category of the related research has focused on deriving narrative from sport games statistics [1, 14, 4]. Such efforts mostly target generating new reporting style narratives about sports games. Apart form the heavy stylistic conformity to the sports commentaries, they also have specifically annotated statistical data about the rules of the game. They include high level statistics about teams' performances, and not necessarily single actions performed by the team members.

3 A Generalized Framework

In this section, we introduce a framework for generating stories about interesting spans of events in a social interaction between a human and an intelligent agent. For this purpose, we have designed and developed a high-level architecture depicted in Fig. 1. It is important to note that this framework is generic, in that it has been designed to be used for new contexts, and create stories from new types of interactions, with minimal change.

We start with raw interaction logs that are output by external interaction systems. These logs contain information about the interaction events, in various non-standardized formats. Such external logs may contain different event



Fig. 1. The high level architecture of the framework.

types and encode them in different ways. In order to be able to standardize and automate our story generation process, we will define and use a notation for describing the interaction events (*AIL*). This notation allows us to perform analyses on the interaction events to find possible deviations from the ordinary event sequences. After which, using a simple natural language generator (SimpleNLG [10]), we produce readable stories about the interesting spans of events in the interaction.

3.1 Processing Input

As noted earlier, the goal in creating a new notation for describing the interaction events is to obtain a standardized format which makes it possible to analyze the event sequences statistically and also makes it easier to generate natural language based on them. We are inspired by simple relational logic which satisfies many of these criteria, while maintaining simplicity and readability. We refer to this notation as *Abstract Interaction Logs* (AIL).

AIL (Abstract Interaction Logs). AIL consist of a series of parameterized events annotated with *event type*. AIL events are categorized within two main groups: *generic* and *context-specific*. Generic events are the ones that can happen in different contexts of interaction, such as *speech* or *facial expression*. Context-specific events are defined for a specific context in which the interaction is occurring, allowing the AIL to capture and leverage such events most efficiently. Below, is the general format of an AIL event (*AILE*) for all types:

AILE (sub, obj, type, content, context, time)

An event is Context-Specific, if the *context* field of the AILE has a value. Furthermore, *type* parameter determines the event type which can be, for instance, a *speech* if the event is generic, or a *meld* in the context of a rummy game. Moreover, *sub* and *obj* denote the subject and the object for an event, who can be any of the characters involved in the interaction. If the *sub* field has a value, then the event is considered to be an *action*. Furthermore, the *content* field allows an event to have information specific to itself, such as an utterance for an speech event, or a card number for a card game move. Lastly, *time* contains the time-stamp for an event. This parameter can be very useful if the domain involves complex temporal sequence.

The more specific and fine-grained types that each event type can have are called *event sub-types*. The event sub-types for the two built-in generic event types that the framework currently supports, are shown in the Table 1 below.

Table 1. The sub-types for the two built-in generic event types. The sub-types list is easily expandable.

Generic Event Type	**Sub-Types**
Speech	Assertives, Directives, Commissives, Expressive, Declaratives
Facial Expression	Smile, Laughter, Gaze, Nod, Eyebrows-up, Head-shake

In order to cover the speech sub-types most efficiently, we have adopted the illocutionary acts from Speech Act theory [16] and the taxonomy introduced in [17]. Below is a small description of the taxonomy.

- **Assertives**: speech acts that commit a speaker to the truth of the expressed proposition
- **Directives**: speech acts that are to cause the hearer to take a particular action
- **Commissives**: speech acts that commit a speaker to some future action
- **Expressive**: speech acts that express the speaker's attitudes and emotions towards the proposition
- **Declaratives**: speech acts that change the reality in accord with the proposition of the declaration

The optimal coverage of the interaction events depends on the sufficient addition of Context Specific events, which make it possible for the AIL to be compatible with interaction logs of different domains. In a robotic task of home cleaning, for example, perhaps moving the objects is a potentially important event that can lead to interesting event sequences, and in an interaction with a social virtual assistant (e.g., in a smartphone), asking for a phone call to be placed could be an informative event.

Characters and meta-data: In addition to the list of AILEs, each AIL also contains a set of *Characters*. Each character includes at least a name and optionally a gender. The most important meta-data for an AIL is an *activity name*, which is used for creating Lead Sentences as explained later. Future work can bring more character specification and meta-data to AIL.

Converting Raw Logs to AIL. The conversion from raw interaction logs to AIL is different for various input logs, and is mainly performed by text processing. After extracting the events, the framework provides easy ways to create

AILE instances with various types and parameters. New AILE types and subtypes can easily be added to the system as mentioned before.

3.2 Detecting Interesting Events

We want to detect limited spans of a given AIL that can represent interesting event sequences in the original interaction. More specifically for this paper, we want to find sequences that are considered memorable and worth telling a story about.

Sequential Rule Mining Algorithm. In order to find the interesting event sequences in an AIL, we have taken a data mining approach, and perform sequential rule mining by treating every AIL in a training set as a sequence of events. Such approach yields a series of *sequential rules*, which, as depicted below, state what set of events (*itemset* S or consequence) are most likely to follow another set of events (*itemset* P or antecedent). Itemsets S and P are unordered and disjoint.

$$\boxed{P = \{e_4, e_1, e_3\} \Longrightarrow S = \{e_2, e_9\}}$$

Training sets are often associated with a specific context. In order to find rare event sequences most efficiently and to capture the contextual information in doing so, we need to find sequential rules that are rare (or common) to several sequences (AILs). To satisfy this requirement, we use the *CMRules* sequential rule mining algorithm [9]. CMRules, which is based on Association Rule Mining, is specifically designed to find sequential rules common to several sequences (a detailed description of CMRules is outside the scope of this paper). The resulted sequential rules are presented with *support* and *confidence* numbers, as shown in the example below. Given an antecedent itemset, these rules could then be used to find least common (or anomalous) event sequences through either leveraging least probable consequence itemsets, or detecting the cases in which the most probable ones are violated.

$$\boxed{P = \{e_4, e_1, e_3\} \Longrightarrow S = \{e_2, e_9\},\ \textit{sup: 0.5, conf: 0.8}}$$

The sequential rule above, for example, states that if events e_4, e_1 and e_3 are seen in any order, with a confidence of 0.8 (*conf*), they will be followed by an e_2, and an e_9 event, in some order. It also states that such rule has a support of 0.5 (*sup*), because it appears in half of the provided event sequences (i.e., half of the AILs).

Therefore, the general sequential rule for our system follows the format below, where P and S are disjoint unordered itemsets of events, and *sup* and *conf* denote *support* and *confidence* as more precisely defined below.

$$\boxed{(P \Longrightarrow S),\ \textit{sup},\ \textit{conf}}$$

- **Support**: the number of sequences that contain $P \cup S$ divided by the number of all sequences.
- **Confidence**: the number of sequences that contain $P \cup S$, divided by the number of sequences that contain P.

Using Sequential Rules to Find Rare Spans. After running the CMRules algorithm on a given set of AILs (as the training set), we obtain a series of sequential rules as described earlier. Our goal is to find rare spans of the interaction (event sequences) that have a higher chance of representing an interesting series of events. To do this, we first consider the sequential rules with a *sup* lower than a specific threshold, which filters our original rule list significantly. We then sort the resulting list by *conf*, and then the *sup* rates so that the rules with lower confidence appear higher in our list. We call this list *Rule List 1*. In a second list of rules, *Rule List 2*, we also extract and sort the most common rules (with highest *sup* and *conf*).

Afterwards, we explore the AILs in our test set and find matching event sequences for the *Rule List 1*, for which both S and P itemsets have occurred according to the rule. We also look for violation cases for rules in *Rule List 2*, for which the S itemset is occurring, but the P itemset is not. Sorted rules of both lists therefore result in a unified sorted list of extracted spans for every AIL and we consider that the spans that appear higher in this list, represent a more interesting sequence for every interaction.

3.3 Generating Natural Language Output

The next step in creating the story is to generate natural language based on the extracted spans of an AIL. In an initial attempt, we perform this by converting each AIL event (AILE) to a single sentence. In this process, we use SimpleNLG [10] which is capable of creating sentences from words and their assigned Part of Speech tags. SimpleNLG also supports changing the verb tense, and the addition of adjectives and propositional phrases.

Relating AILEs to Sentence Generation. In order to create a sentence from an AILE, we firstly need a verb. To yield one, for every AILE sub-type, we keep a verb stem that most closely describes the event or action. After which, we use the story characters that are assigned to each AILE as *sub* and *obj*, and since each character contains a name string, we can obtain name strings for subject and object, where applicable. Moreover, using the SimpleNLG's capability to change a verb's tense, we can use the *time* of each AILE to determine a verb tense, when compared to a given narration time (usually the present moment). Lastly, we use the AILE *content* parameter as a propositional phrases for our sentence generator, to add to the sentence type specific additions, such as quotations for a speech action.

To summarize, the conversion between an AILE's parameters and the tokens required to build a sentence is shown in Table 2.

Table 2. The mapping between AILE parameters and sentence tokens

AILE Parameter	Sentence Tokens
Sub-type (or *type*)	Verb
Sub character's name	Subject
Obj character's name	Object
Content	Propositional-phrase
Time	Verb tense

Adding Lead Sentences to Story Spans. In order to provide a context for the story spans, if it were to be presented to the reader separately, the framework builds a single "lead sentence" to appear before a story span, which plays a role similar to the role of an *Orientation* in Labov-Waletzky model [13]. Lead sentences briefly inform the listener about the type of activity in the interaction. For a context-specific set of events, we use the activity name variable of an AIL (e.g., *playing Rummy*), and for a generic set of events, if available, we use the time of the day in which the events have been happening (e.g., *In an afternoon*). In conjunction with character names, which are provided as meta-data in the AIL, lead sentences are shaped as a full sentence describing the interaction as an activity (e.g., *"During house cleaning by Jim and Karen"* or *"Once upon a time in an afternoon"*). If the none of the required meta-data are available, a leading phrase could simply be *"once upon a time"*.

4 Study

In order to evaluate our approach, we applied this framework to a specific context of social game-playing. As mentioned before, we had access to interaction logs from social gameplays between participants and a social virtual agent from a work by [2], with the related interface shown in Fig. 2. In the following sections, we will describe the specific interaction logs and context-specific events used for this study, explain our subjective evaluation, and provide and discuss our results.



Fig. 2. The interface of the system that generated the logs used in our study. Users played a social games of rummy with a virtual agent.

4.1 Raw Logs

The gameplay logs database contains 15 cases of interaction logs happening during rummy games with a social virtual agent capable of making verbal comments about the game and showing facial expressions that relate to the semantics in the game. The logs contain time-stamped records of the following events:

- User's game move (Draw, Discard, Layoff, Meld)
- Agent's game move (Draw, Discard, Layoff, Meld)
- User's smile and laughter
- User's comment on a move or response to an agent's comment
- Agent's comments on a move or response to a user's comment

Such events are translated into generic and context-specific AILEs seen in Table 3 below.

Table 3. Generic and context-specific AILEs in rummy logs. "FE" stands for Facial Expression. Sub-types, where applicable, are shown in parentheses.

Type	**AILEs**
Generic	Speech (expressive), FE (smile), FE (laughter)
Context-Specific	Draw, Discard, Meld, Layoff

A sample of the AIL for a rummy game is shown in Table 4. Moreover, Table 5 shows a full rummy story, consisting of all the events in the interaction, while Table 6 shows the interesting spans of the same story, picked up by our framework. In the first interesting span, we see that the agent (Karen) seems to be on a streak of strong moves, and the user is complementing Karen. The second span covers the events of a surprising game ending, in which Karen first brags about finishing her cards after playing a layoff move. Afterwards, the user responds to that by saying "not so soon", and plays a finishing meld move to win the game. The user then reacts to the situation by smiling and talking about it.

4.2 Procedure

To evaluate our approach, we conducted a pilot study in which 11 participants were asked to choose the more interesting story span between two options. These two options consisted of an interesting story span generated by the framework, and a non-overlapping random span of the same story with an equal length, as a baseline. Each participant was presented with an identical set of 8 questions, while the ordering of the questions and their answers was randomized.

4.3 Results

Out of 88 choices made in 8 questions by 11 participants, in 62 cases, the system generated interesting story spans were favored by the participants, while in 26

Table 4. A sample of a rummy AIL; Null values are used when a parameter is not applicable.

LayOffEventAILE(Karen, card, null, null , rummy, 11:15:50) DiscardEventAILE(Karen, card, null, null , rummy, 11:15:56) SpeechAILE(Karen, User, EXPRESSIVE, yes, i am finishing my cards, null, 11:16:03) ResponseAILE(User, Karen, EXPRESSIVE, not so soon, rummy, 11:16:04) MeldEventAILE(User, card, null, null , rummy, 11:16:17) FacialExpressionAILE(User, null, SMILE, null , null, 11:16:19) GameOverAILE(game, null, null, null , rummy, 11:16:27) SpeechAILE(User, Karen, EXPRESSIVE, i finally win. good game., null, 11:16:27) FacialExpressionAILE(User, null, LAUGHTER, null , null, 11:16:28) FacialExpressionAILE(User, null, SMILE, null , null, 11:16:29) FacialExpressionAILE(User, null, LAUGHTER, null , null, 11:16:29)

Table 5. A sample full story generated for a rummy interaction

There once was an agent named Karen who was playing rummy with a human named user. Karen melded card. Next, Karen discarded card. Karen told User how's your hand over there?. Afterwards, User responded to Karen good!. User discarded card. Subsequently, Karen told User i should say, you do play very well. User responded to Karen thank you, you do too. Karen discarded card. User laughed. Next, User smiled. After that, User laughed. After which, User discarded card. Karen discarded card. After that, User stopped laughing. User melded card. User laughed. User smiled. Afterwards, User discarded card. Karen laid off card. Karen discarded card. User told Karen nice. After which, User smiled. After which, User discarded card. Karen laid off card. Karen discarded card. User told Karen nice again. User laid off card. After that, User discarded card. User told Karen found a lay off, yes!. Karen responded to User. After that, Karen discarded card. After which, Karen told User i am in this to win. Afterwards, User responded to Karen well do your best. User discarded card. After that, Karen discarded card. User told Karen you do play very well. Karen responded to User good job. User discarded card. Karen told User humans are somewhat intelligent but robots are genius. User responded to Karen bragging won't get you to win madam agent. Afterwards, Karen melded card. Next, Karen discarded card. Afterwards, User responded to Karen good meld. Next, Karen responded to User such encouragement to hear that!. User melded card. Afterwards, User discarded card. User told Karen only few cards left for you. Afterwards, Karen responded to User yes, i just found noticed myself. Karen discarded card. Afterwards, Karen told User i am in this to win!. Next, User responded to Karen well do your best. User laid off card. Then, User discarded card. Then, Karen discarded card. Afterwards, User told Karen you do play very well. Afterwards, Karen responded to User i know, i am excited. Subsequently, User discarded card. Karen laid off card. Karen discarded card. Next, Karen told User yes, i am finishing my cards. User responded to Karen not so soon. Subsequently, User melded card. Afterwards, User smiled. Game ended. Afterwards, User told Karen i finally win. good game. User laughed. User smiled. User laughed. Next, User smiled. After which, User laughed.

cases, random spans were favored. Therefore, in 70% of the cases, participants identified the story spans generated by the system as more interesting than a random baseline.

Table 7 provides the detailed number of cases in which participants agreed or disagreed with the system in each of the 8 questions in our questionnaire, along with their respective p-values from a one-tailed binomial test.

We also calculated the inner-agreement of our participants, which yielded a Krippendorff's alpha [12] of 0.32.

5 Conclusions and Future Work

The results of the pilot study suggest that in almost 3 out of 4 cases, study participants prefer system generated output over randomly selected span of events

Table 6. Two interesting spans generated by our framework from the full story in Table 5.

(1) During Karen and User's rummy game, this happened: Karen laid off card. Then, Karen discarded card. After which, User told Karen nice. Afterwards, User smiled. User discarded card. Afterwards, Karen laid off card. Karen discarded card. User told Karen nice again. **(2)** During Karen and User's rummy game, this happened: Karen laid off card. Then, Karen discarded card. Next, Karen told User yes, i am finishing my cards. Subsequently, User responded to Karen not so soon. User melded card. User smiled. Game ended. User told Karen I finally win. good game.

Table 7. Questionnaire results for every item, including a p-value from a binomial test.

Questions	Agreements out of 11	p-value
q1	2	**0.03**
q2	4	0.27
q3	10	**<0.006**
q4	8	0.11
q5	11	**<0.001**
q6	11	**<0.001**
q7	7	0.27
q8	9	**0.03**

for interestingness. Despite the limited number of participants, these results are statistically significant in 5 of our 8 questionnaire items. This is particularly promising when considering the fact that the raw interaction logs we used in this study had a very limited variety in terms of event types. We think that this pilot study provides initial validation of the framework.

An immediate future work for us would be to expand our mining algorithm to leverage information beyond just the event types, to include event parameters from each AILE. Not unrelated to such addition, would be to also consider a context-dependent multi-dimensional space that covers the semantics of an AIL's context, through a number of variables (such as "move strength" and "move novelty" in a game), in which every AILE is assigned a single point. Such covering of a context's event semantics can result in finding extraordinary event sequences that are not necessarily or merely rare, but are also semantically interesting.

It would also be interesting to see if adding a story representation system, such as SIG [6], would help to enrich the interaction dynamics that our system can support, and effectively add goals, affect and other interpretative artifacts to the AIL notation. It is also very important to choose the right moments and "trigger events" that motivate telling stories about prior interactions. Such semantic story representation for event sequences can also enable a search among interesting stories of prior interactions that are semantically analogous to a newly observed event sequence.

Additionally, we can change the *telling* of the story, which can be achieved using tools such as PERSONAGE [15] or new tools inspired by it. Moreover, we might benefit from leveraging information about users' emotional state to inform different stages of this process, such as the selection of interesting event sequences, telling of the story, or triggering of one.

Furthermore, we imagine a more sophisticated Natural Language Generation engine to be very advantageous, especially in the evaluation phase. Last but not least, a *collaborative retelling* of interesting stories will allow an even higher level of interactivity and completes a scenario in which an intelligent agent and its companion, collaboratively tell interesting stories about the experiences they have shared, to a third being.

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